

# Describing Admissible Strings I

Formal grammars, context-free grammars, and linear-indexed grammars

Cem Bozsahin

LING487 · Fall 2026

# A grammar is a finite description of a stringset

## Question

How can finitely many rules characterize arbitrarily many strings?

$$G = (V, \Sigma, R, S)$$

- $V$ : nonterminals;  $\Sigma$ : terminals.
- $R$ : rewrite rules;  $S$ : start symbol.
- $\mathcal{L}(G) = \left\{ w \in \Sigma^* \mid S \xRightarrow{*} w \right\}.$

MOLR §§4.1–4.2

# Where we are

1. Formal grammars
2. Context-free grammars
3. Linear-indexed grammars

# Rule form determines descriptive power

---

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| <b>Regular</b>           | $A \rightarrow wB$ or $A \rightarrow w$ (one-sided nonterminal) |
| <b>Context-free</b>      | $A \rightarrow \beta$ (one nonterminal on the left)             |
| <b>Context-sensitive</b> | $\alpha \rightarrow \beta$ , with context and noncontraction    |
| <b>Unrestricted</b>      | $\alpha \rightarrow \beta$ , $ \alpha  > 0$                     |

---

## Take-away

A grammar family is a restriction on descriptions, not merely on the strings already observed.

# Derivation is a relation over intermediate strings

For

$$S \rightarrow aSb \mid ab,$$

one derivation is

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaabbb.$$

- $\Rightarrow$  records one licensed rewrite.
- $\Rightarrow^*$  records zero or more steps.
- The grammar describes every terminal string reachable from  $S$ .

## Guided analysis

Derive  $aabb$ . Can this grammar derive  $abab$ ?

# Where we are

1. Formal grammars
2. Context-free grammars
3. Linear-indexed grammars

# One continuing nonterminal can coordinate two edges

$$S \rightarrow AB, \quad A \rightarrow aAb \mid ab, \quad B \rightarrow cBd \mid cd$$

$$\mathcal{L}(G) = \{a^n b^n c^m d^m \mid n, m \geq 1\}.$$

- $A$  coordinates the numbers of  $a$ 's and  $b$ 's.
- $B$  independently coordinates  $c$ 's and  $d$ 's.
- The derivation tree records constituent containment.

MOLR §4.1

# The same stringset can receive different structures

## Weak equivalence

Two grammars generate the same set of terminal strings.

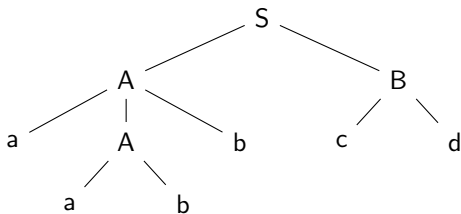
## Strong equivalence

They also assign the relevantly same structures.

## Guided analysis

Why can two weakly equivalent grammars make different linguistic predictions?

# A derivation tree is not merely a record of rewrite order



- Dominance and constituency remain after incidental rewrite order is ignored.
- The yield restores the terminal string from left to right.
- Ambiguity occurs when one string has more than one relevant structure.

# Where we are

1. Formal grammars
2. Context-free grammars
3. Linear-indexed grammars

# Indexes pass a controlled dependency through a derivation

- A nonterminal may carry a stack of index symbols.
- Rules push, pass, or pop indexes along one distinguished path.
- The mechanism extends context-free coordination without unrestricted copying.

$$A[\gamma] \Rightarrow \dots \Rightarrow B[\gamma f] \Rightarrow \dots$$

## Take-away

The extra memory is structurally localized rather than globally available.

MOLR §4.2

# Mild context-sensitivity targets limited cross-serial dependency

- Some natural-language dependencies exceed ordinary context-free descriptions.
- They remain far below unrestricted rewriting.
- Linear-indexed descriptions preserve polynomial parsability and constrained growth.

## Question

Why should a linguist prefer the weakest formalism that preserves the observed dependency?

## Third hour: grammar as a claim

Given  $G$  with  $S \rightarrow aSb \mid \epsilon$ :

- 1 State  $V, \Sigma, R, S$ .
- 2 Derive  $aabb$ .
- 3 State  $\mathcal{L}(G)$  in set-builder notation.
- 4 Identify one invariant of every generated string.
- 5 Say what evidence would make this grammar linguistically inadequate despite weak correctness.

## Week 6 synthesis

- ① A grammar is a finite description, not the language itself.
- ② Rule restrictions define grammar families.
- ③ Derivations license strings; trees preserve structural relations.
- ④ Weak equivalence does not guarantee structural equivalence.
- ⑤ Linear indexes extend memory along a constrained dependency path.

### Question

Quiz 5 asks for a tiny grammar, derivation, or interpretation of one.